1 DMC Implementation Guide - Factory Model Editor

Purpose

This document describes how to operate the Factory Model Editor (FME).

Requirements

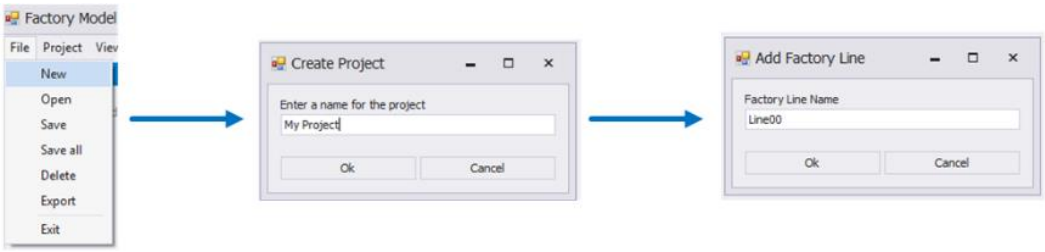
License: DMC-MDL

MPDV-DMC is required.

Designing a factory model in the DME-FME

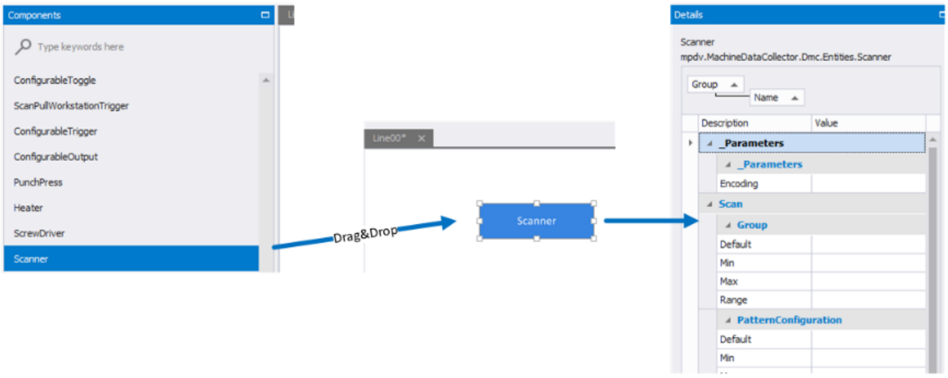
The main function of the DME-FME is the graphic modelling of a factory model for the DMC. The following sections illustrate how to design a factory model.

First create a new FME project including a *factory line.* To create a new project, click "File“ → "New“. Assign a unique project name. Then create a *factory line* by indicating a unique name and clicking "Ok". A new project has been created. The *factory line* now appears as an empty tab on the FME workspace.



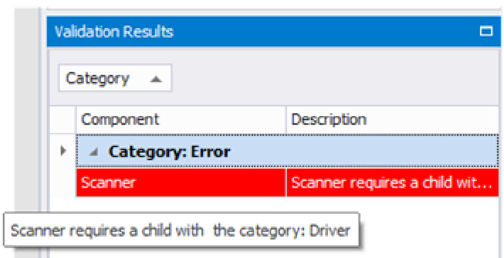
Now start modelling your *factory line.* Go to the "components" and select a component required for your *factory line.* If required, use the search field at the top of the screen to search for the required component. If supported by the component, the component shows additional information on properties and requirements if you hold the mouse pointer over a component (mouseover). Once you have selected an appropriate component, drag and drop this component in the workspace of the *factory line.*

You can now configure the component you have added to the workspace.



Click and select the component you want to configure directly in the workspace. The "details" view shows the available parameters with a description and a value. You can directly edit these parameters in the "details" view. To do so, select the required parameter and enter its value in the "value" column. You can also add further parameters to a component. In the "details" view, click the button "add parameter" and enter the parameter's name and value. When adding parameters, you can choose to create custom parameters or data items. Observe the naming convention for data items (names in square brackets). Click the button "remove selected parameter" in the "details" view to delete a parameter. Please note that you can only delete the parameters you created. After editing all parameters, you can add further components to your model. Connect the components of your workspace appropriately in order to complete data modeling. Connect components to shape their parent-child relationship. Open the context menu of the first component (by right-clicking the component) to model the parent-child relationship. Select the option "connect". Then click the component you want to connect (target).

Check the results of component validation in "validation results" to complete data modeling.



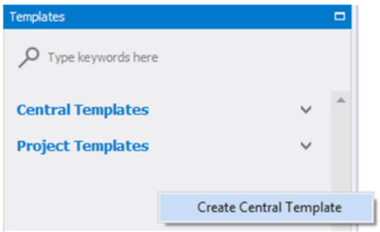
Every time you change a model, the system validates the components and checks their dependencies and requirements. The "validation results" section highlights errors in red. Double click an entry to focus the relevant component in the workspace. The column "description" shows a description of the error. Correct the errors. Then data modeling is completed.

Create and use templates

The FME also provides templates to model a *factory line.* We distinguish between *central templates* and *project templates.* They differ in how they can be reused. You can use central templates for all projects. But you can use project templates only in the current project.

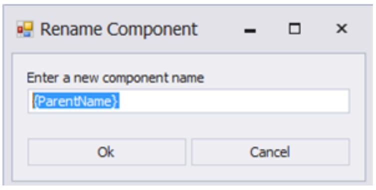
Create central template

The following paragraphs describe how to create a central template in the FME.



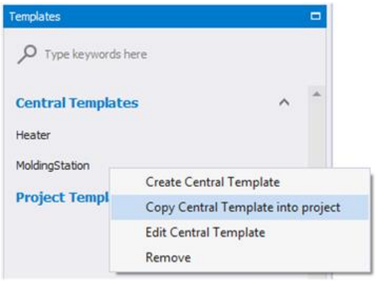
Creating a central template:
Go to the detail application "templates"**.**
Right-click to open the context menu for central templates.
In the context menu click the option "create central template".
Assign a unique template name.
Confirm the name by clicking "Ok". An empty tab opens in the workspace to create the central template.
Now select the components to create the template like designing a *factory model.*

You can preconfigure the components and also assign placeholders for names and parameters. When you use the template for the first time (when you add the template to the workspace), you can adjust the placeholders via an editing dialog. To assign placeholders, enter the name of the required value in braces (example: "**{**ParentName**}**“).



Create project templates

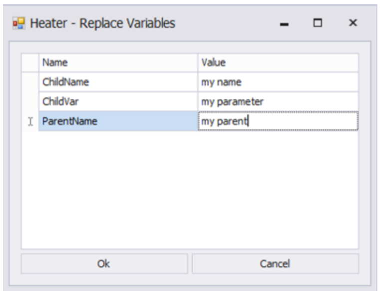
Use a central template to create a project template. Proceed as follows to create a project template:



Create a central template (as described in section "create central template") or select an existing template.
Right-click this template.
In the context menu click the option "copy central template into project".
Then this template is available as a project template.

Use templates

If you use central templates, you have to make them available as project templates at first (see "create project templates"). To use project templates, drag and drop them in the workspace. If the template includes placeholders, a dialog opens where you can edit the placeholders.



The workspace highlights added template components in green. In general, you cannot edit templates directly. But you can cancel the connection between templates and components. Then the components become "standard" components. Then you can edit these components as usual. To edit an inserted template, right-click the template (green) in the workspace. In the context menu, click the option "break template instance".
The function "break template instance" cancels the connection between the components and the template and you can edit the components as usual.